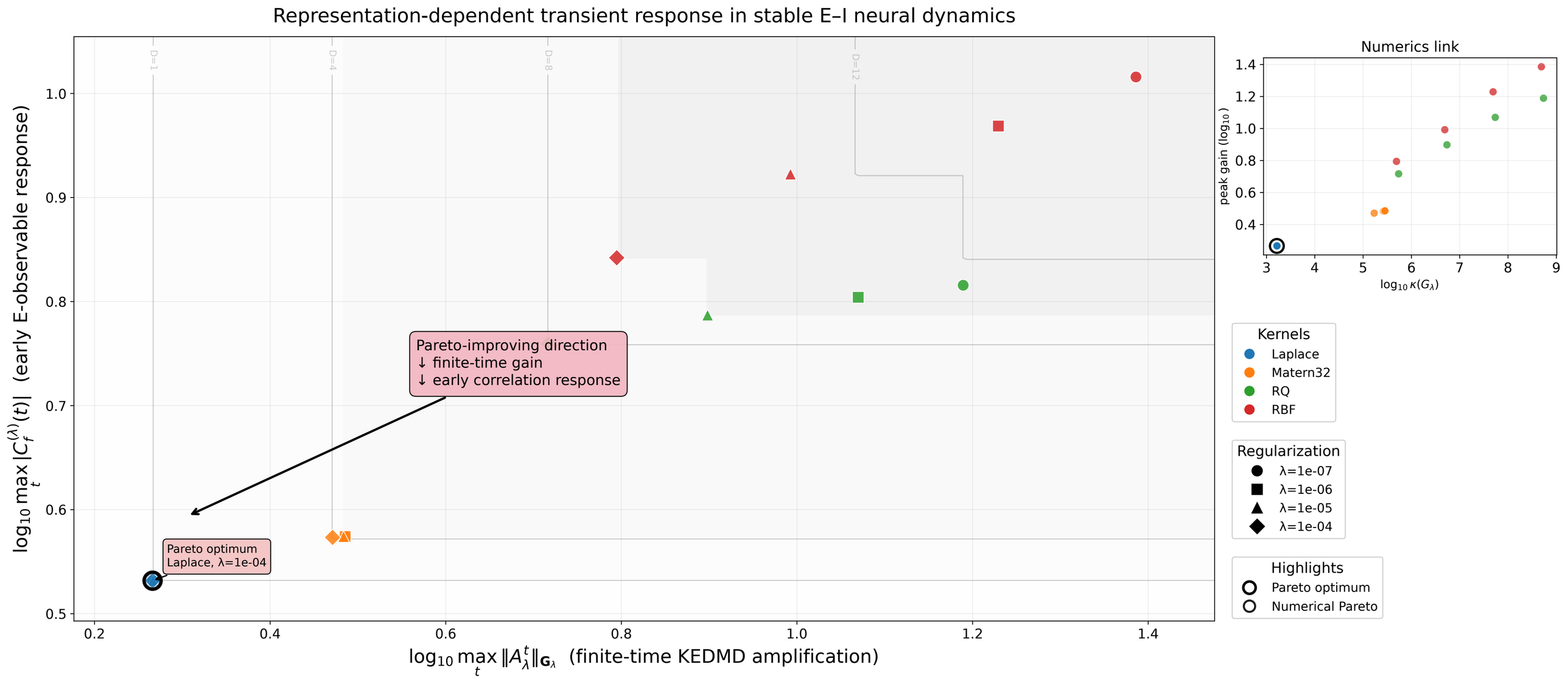

Representation-dependent transient response in stable E–I neural dynamics
D=1
D=4
D=8
D=12
Pareto-improving direction
↓ finite-time gain
↓ early correlation response
Pareto optimum
Laplace, λ=1e-04
$\log_{10} \max_t |C_f^{(\lambda)}(t)|$ (early E-observable response)
$\log_{10} \max_t \|A_\lambda^t\|_{\mathbf{G}_\lambda}$ (finite-time KEDMD amplification)
Numerics link
peak gain ($\log_{10}$)
$\log_{10} \kappa(G_\lambda)$
Kernels
Laplace
Matern32
RQ
RBF
Regularization
λ=1e-07
λ=1e-06
λ=1e-05
λ=1e-04
Highlights
Pareto optimum
Numerical Pareto